Which of the following sentences is true (at least according to Forbes)?

Without sufficient data, tacit knowledge is useless in practice

Knowledge graphs are the only way to make tacit knowledge explicit

There is usually more time spent on preparing, linking and cleaning data than on building the datasets in the first place

The more different datasets need to be integrated, the more suitable relational databases are

Convert to prefix syntax: ¬((A → (¬B ∨ C)) v (¬D v ¬E))

5

¬(v(→(v(v(v(¬(¬(¬(A, B,C,D,E))))))

0

((((E)¬,(D)¬)v,((C,(B)¬)v,)→)v)¬

8

¬(v(→ (E,v(¬(B),D)),v(¬(C),¬(A))))

72

¬(v(→ (A,v(¬(B),C)),v(¬(D),¬(E))))

Consider the following propositional formula: $p \rightarrow (\neg q \wedge r)$. Which of the following interpretations is a model?

2

I(p)=True, I(q)=True, I(r)=True

77

I(p)=True, I(q)=False, I(r)=True

1

I(p)=True, I(q)=False, I(r)=False

4

There is no model

84

Let KG={(a,p,b), (b,q,a)} be a knowledge graph. In Simple KG logic, which of the following interpretations is a model of KG?

4

(U,I) with U={1,2}, I(a)=1,I(b)=2, I(p)={(1,2)}, I(q)={(1, 1)}

4

(U,I) with U={1,2}, I(a)=2,I(b)=2, I(p)={(1,2)}, I(q)={(2,2)}

70

(U,I) with U={1,2}, I(a)=2,I(b)=1, I(p)={(2,1)}, I(q)={(1, 2)}

4

(U,I) with U={1,2}, I(a)=2,I(b)=1, I(p)={(2,1)}, I(q)={(2, 2)}

Suppose you want to convert this table to a KG. Determine the Literals, Resources and Properties (one correct answer per type)

Student	First Name	Last Name	Birthdate	University
s1	Maria	Dienst	11-08-1993	uni1
s2	Adriana	Kaat	13-01-1996	uni1
s3	Peter	Janssen	02-03-1994	uni2

University	Acronym	City	Country
uni1	VU	Amsterdam	Netherlands
uni2		Paris	France

11

Resources: s1, Maria, uni1, Amsterdam, 11-08-1993

✗

73

Resources: s2, uni1, Amsterdam, Netherlands

✓

72

Literals: Maria, Dienst, 11-08-1993

✓

12

Literals: Adriana, Kaat, uni1

✗

3

Properties: studiesIn, locatedInCountry, uni2

✗

81

Properties: hasFirstName, hasLastName, hasAcronym

✓

Which of the following properties can be safely modelled as an instance ofowl:InverseFunctionalProperty?

2
ex:hasFirstName

6
ex:hasLastName

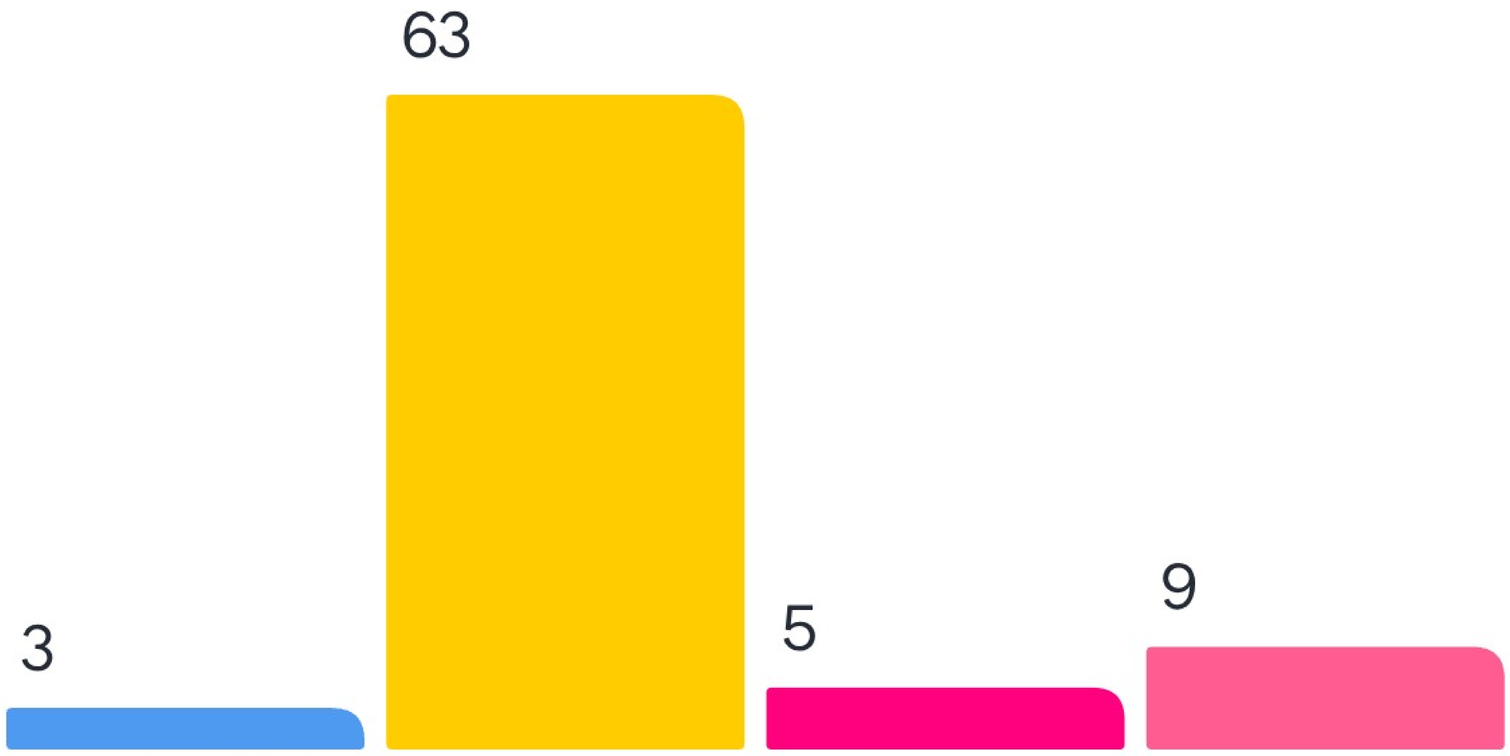
7
ex:hasBirthDate

65
None of the Above

80

Explain what the SPARQL query below tries to achieve (open question)

```
SELECT ?x (COUNT(?y) as ?nb_y)
WHERE {
?y <http://example.org/studiesIn> ?x
} GROUP BY (?x) ORDER BY DESC (?nb_y)
```



Return a list of people that study in a specfiic city

Return a count of people studying in a city, ordered by count

Return a table with three columns, x, y and nb_y, of who studies where how many times

Return a list of properties that are subproberties of studiesIn, ordered by count

80

Consider the following OWL ontology. Which of the following statements can be derived?

```
ex:EuropeanCitizen owl:equivalentClass [ rdf:type owl:Restriction ;
                                         owl:onProperty ex:citizenOf ;
                                         owl:allValuesFrom ex:EuropeanCountry ].
ex:john ex:citizenOf ex:usa.
ex:john rdf:type ex:EuropeanCitizen.
```

5

ex:john rdf:type ex:American

7

ex:usa owl:sameAs ex:EuropeanCitizen

30

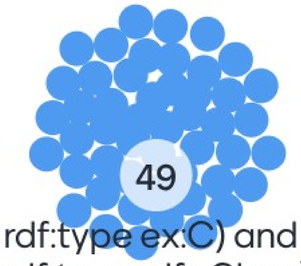
ex:usa rdf:type ex:EuropeanCountry

34

Nothing, as the ontology is inconsistent

KG= {(ex:s ex:p ex:o), (ex:p rdfs:domain ex:C), (rdfs:domain rdfs:range rdfs:Class), (ex:q rdf:type ex:B)}. Which triples are derived using rules 2+3

rdfs2	**aaa rdfs:domain XXX. yyy aaa zzz.**	**yyy rdf:type XXX.**
rdfs3	**aaa rdfs:range XXX. yyy aaa zzz.**	**zzz rdf:type XXX.**



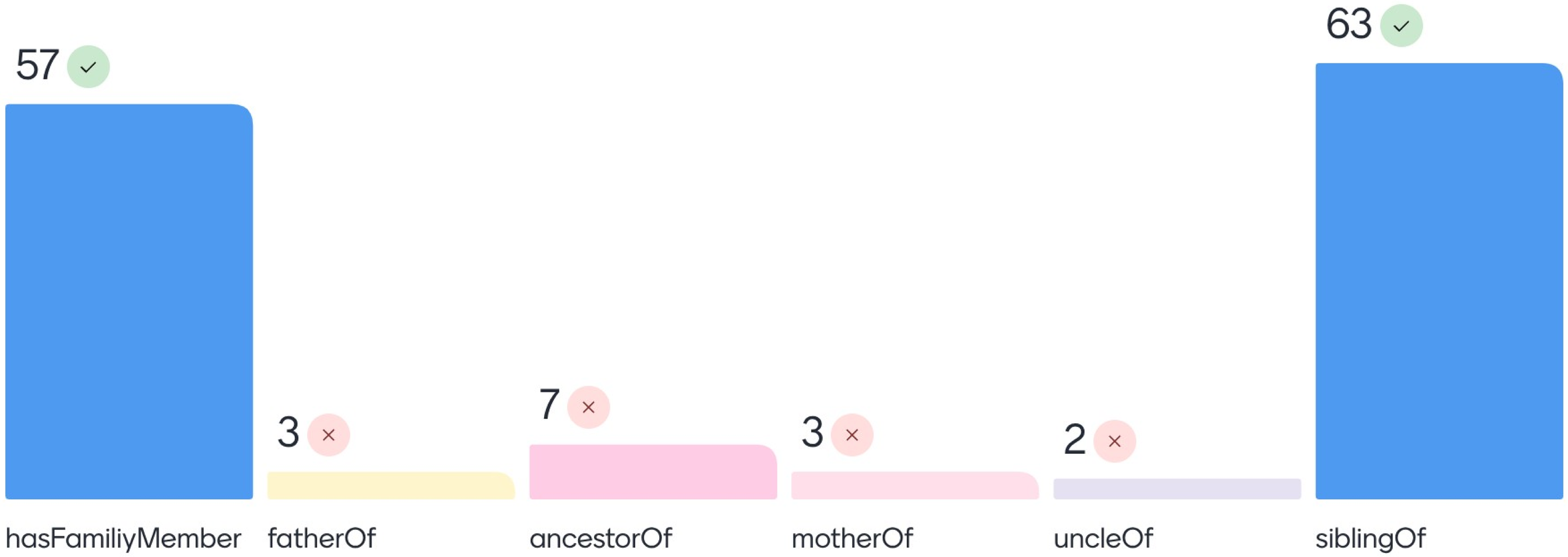
(ex:s rdf:type ex:C) and (ex:C rdf:type rdfs:Class)

(ex:o rdf:type ex:C) and (ex:C rdf:type rdfs:Class)

(ex:s rdf:type ex:B) and (ex:p rdf:type rdfs:Class)

(ex:o rdf:type ex:B) and (ex:C rdf:type rdfs:Class)

Which of the following relations is both symmetric and irreflexive? (mult. select)



74

Consider the following OWL ontology. Which of the following statements can be derived?

```
ex:EuropeanCitizen rdfs:subClassOf ex:Person ;
ex:EuropeanCitizen rdfs:subClassOf [ rdf:type owl:Restriction ;
                                     owl:onProperty ex:citizenOf ;
                                     owl:someValuesFrom ex:EuropeanCountry ].
ex:john ex:citizenOf ex:netherlands.
ex:netherlands rdf:type ex:EuropeanCountry.
```

3

ex:john rdf:type ex:EuropeanCountry

27

ex:john rdf:type ex:EuropeanCitizen

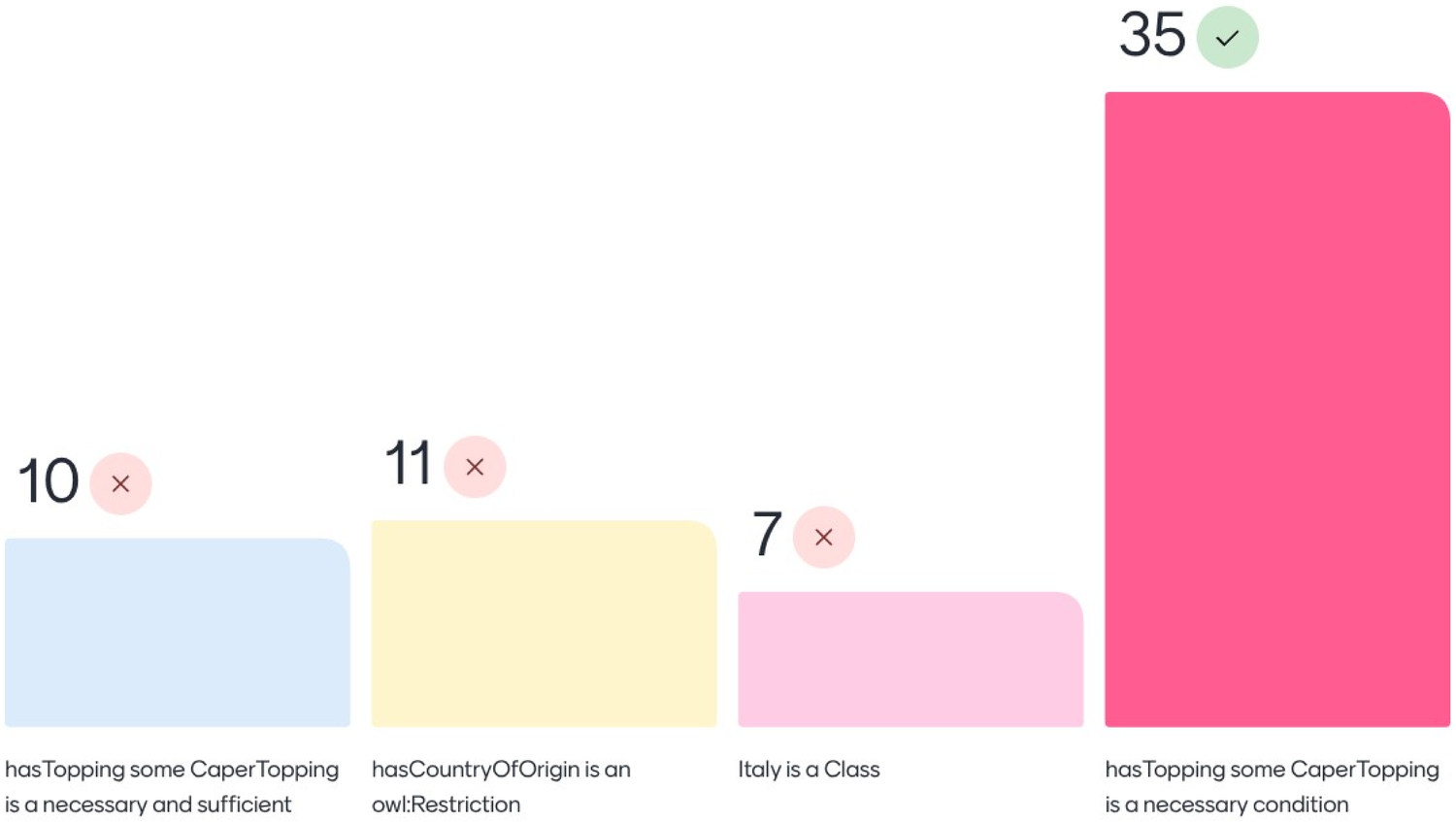
18

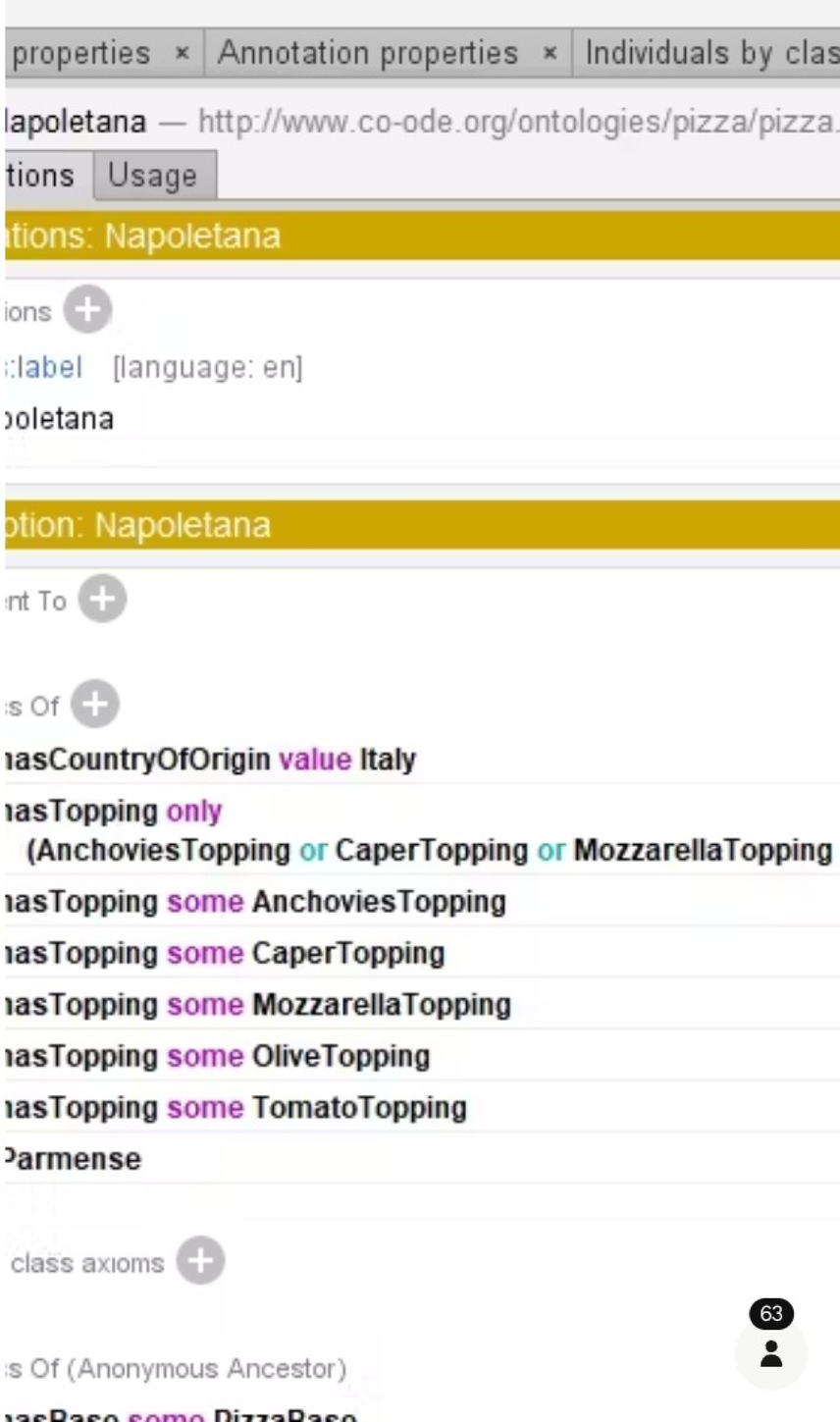
ex:john rdf:type ex:Person

17

None of the above

Which of the following statements is true





Which of the following statements are true? (mult. select)

9

To express hierarchies in a taxonomy, OWL is needed

28

A taxonomy only contains terms that are organized in a hierarchical structure

13

An ontology is more expressive than a taxonomy

6

In formal logic terms, SKOS is more expressive then OWL

56

Given the four following terms: goat, goat_milk, farm, and mammal. Which of the following statements are true? (mult. select)

1

ex:goat skos:broader ex:goat_milk

7

ex:mammal skos:narrower ex:goat

0

ex:goat_milk rdfs:subClassOf ex:goat

0

ex:goat skos:broader ex:farm

29

ex:goat rdfs:subClassOf ex:mammal

0

ex:goat rdf:type ex:farm

0

ex:mammal rdfs:subClassOf ex:farm